MIL-C-11015/26B
28 September 1970
SUPERSEDING
MIL-C-11015/26A
12 August 1969

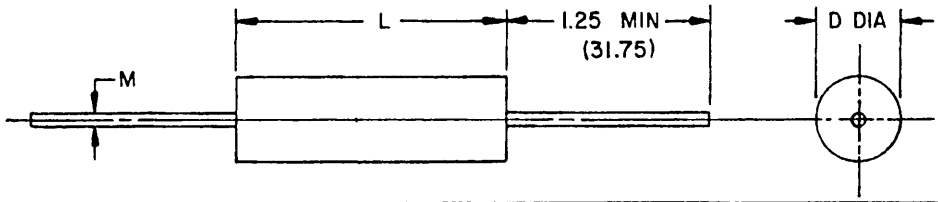
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), STYLES CK18 AND CK19

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN
after 28 September 1970. Use MIL-C-39014/13



Style	Dimension		
	L	D	M (dia)
CK18	.275 (6.99) $^{+.015\ (.38)}_{-.030\ (.76)}$	.090 (2.29) ±.010 (.25)	.020 (.51) $^{+.004\ (.10)}_{-.001\ (.03)}$
CK19	.375 (9.53) $^{+.025\ (.64)}_{-.015\ (.38)}$	.130 (3.30) $^{+.025\ (.64)}_{-.010\ (.25)}$	.020 (.51) $^{+.007\ (.18)}_{-.001\ (.03)}$

NOTES:
1. All dimensions in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See above figure.
Case type - Tubular with axial leads.
Material - Glass or ceramic.
Seal type - Hermetic.
Capacitance value - See table I.
Capacitance tolerance - ±10 percent (K) or ±20 percent (M) as shown in table I.
Rated temperature - -55° to +125°C.

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage - 250 percent of rated voltage.
Body insulation: Not applicable.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100, 000 ft).
Test potential - 100 percent of rated voltage.

Insulation resistance (IR): Method 302 of MIL-STD-202. 100, 000 megohms, min, or 1, 000 megohm-microfarads, min, whichever is less.
Test voltage - 100 volts dc.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency:
100 pF or less - 1 MHz ± 1 kHz.
Over 100 pF - 1 kHz ± 100 Hz.
Test voltage - 1.0 ± 0.2 volt rms.

Dissipation factor (DF): 2.0 percent, max.

FSC 5910

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TABLE I. Styles CK18 and CK19 characteristics.

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pF	
CK18BX100-	100	BX	10	K, M
CK18BX150-	100	BX	15	K, M
CK18BX220-	100	BX	22	K, M
CK18BX330-	100	BX	33	K, M
CK18BX470-	100	BX	47	K, M
CK18BX680-	100	BX	68	K, M
CK18BX101-	100	BX	100	K, M
CK18BX151-	100	BX	150	K, M
CK18BX221-	100	BX	220	K, M
CK18BX331-	100	BX	330	K, M
CK18BX471-	100	BX	470	K, M
CK18BX681-	100	BX	680	K, M
CK18BX102-	100	BX	1,000	K, M
CK18BX152-	100	BX	1,500	K, M
CK18BX222-	100	BX	2,200	K, M
CK18BX272K	100	BX	2,700	K
CK18BX332-	100	BX	3,300	K, M
CK18BX392K	100	BX	3,900	K
CK18BX472-	100	BX	4,700	K, M
CK18BX562K	100	BX	5,600	K
CK18BX682-	100	BX	6,800	K, M
CK18BX822K	100	BX	8,200	K
CK18BX103-	100	BX	10,000	K, M
CK19BX153-	100	BX	15,000	K, M
CK19BX223-	100	BX	22,000	K, M
CK19BX273K	100	BX	27,000	K
CK19BX333-	100	BX	33,000	K, M
CK19BX393K	100	BX	39,000	K
CK19BX473-	100	BX	47,000	K, M
CK19BX563K	100	BX	56,000	K
CK19BX683-	100	BX	68,000	K, M
CK19BX823K	100	BX	82,000	K
CK19BX104-	100	BX	100,000	K, M

1/ Where applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance.

Seal: Method 112 of MIL-STD-202, condition C, procedure III; nominal leak rate sensitivity 10^{-8} atm cc/sec. For gross leak test, a dye penetrant test (with comparable leak rate sensitivity) may be substituted for the bubble test.

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Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 G's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

Temperature cycling and immersion:
Temperature cycling: Method 102 of MIL-STD-202, condition D, except step 3 shall be at +125°C.
Immersion: Method 104 of MIL-STD-202, condition B.
IR - 100,000 megohms, min, or 1,000 megohm-microfarads, whichever is less.

Salt spray (corrosion): Not applicable.

Terminal strength: In accordance with MIL-C-11015, 5 pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.
DWV - 250 percent of rated voltage.
IR - 100,000 megohms, min, or 1,000 megohm-microfarads, whichever is less.
Cap. - Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202. 2 terminals.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
IR - 100,000 megohms, min, or 1,000 megohm-microfarads, whichever is less.
ΔCap. - ±5 percent of initial measured value.
ΔDF - 0.5 percent, max.

Voltage-temperature limits: In accordance with MIL-C-11015.

Life (at elevated ambient temperature):
Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).
Test potential - 200 percent of rated voltage.
IR - Not less than 50 percent of initial requirement (at 125°C and 25°C).
ΔCap. - ±1 percent (10 thru 100 pF units) or +10, -15 percent (over 100 pF units) of initial measurement (at 25°C).
DF - 2.5 percent, max (at 25°C).

Marking: In accordance with MIL-C-11015.

Inactivated styles: For new design it is recommended that units of MIL-C-39014/13 be considered as listed in table II. Although table II lists only the highest failure rate level, lower failure rate levels may be considered. For substitutability of the inactivated styles, capacitors in MIL-C-39014/13 of the same capacitance value, tolerance, rated voltage and temperature coefficient can be used regardless of the failure rate designation.

TABLE II. Cross reference of inactivated styles.

From inactivated MIL-C-11015	To MIL-C-39014/13 L (2.0) M39014/13-	From inactivated MIL-C-11015	To MIL-C-39014/13 L (2.0) M39014/13-
CK18BX100K	0001	CK18BX151M	0016
CK18BX100M	0002	CK18BX221K	0017
CK18BX150K	0003	CK18BX221M	0018
CK18BX150M	0004	CK18BX331K	0019
CK18BX220K	0005	CK18BX331M	0020
CK18BX220M	0006	CK18BX471K	0021
CK18BX330K	0007	CK18BX471M	0022
CK18BX330M	0008	CK18BX681K	0023
CK18BX470K	0009	CK18BX681M	0024
CK18BX470M	0010	CK18BX102K	0025
CK18BX680K	0011	CK18BX102M	0026
CK18BX680M	0012	CK18BX152K	0027
CK18BX101K	0013	CK18BX152M	0028
CK18BX101M	0014	CK18BX222K	0029
CK18BX151K	0015	CK18BX222M	0030

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TABLE II. Cross reference of inactivated styles - Continued.

From inactivated MIL-C-11015	To MIL-C-39014/13 L (2.0) M39014/13-	From inactivated MIL-C-11015	To MIL-C-39014/13 L (2.0) M39014/13-
CK18BX272K	0031	CK19BX223M	0046
CK18BX332K	0032	CK19BX273K	0047
CK18BX332M	0033	CK19BX333K	0048
CK18BX392K	0034	CK19BX333M	0049
CK18BX472K	0035	CK19BX393K	0050
CK18BX472M	0036	CK19BX473K	0051
CK18BX562K	0037	CK19BX473M	0052
CK18BX682K	0038	CK19BX563K	0053
CK18BX682M	0039	CK19BX683K	0054
CK18BX822K	0040	CK19BX683M	0055
CK18BX103K	0041	CK19BX823K	0056
CK18BX103M	0042	CK19BX104K	0057
CK19BX153K	0043	CK19BX104M	0058
CK19BX153M	0044		
CK19BX223K	0045		

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MI
Navy - AS, OS, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-1003-5)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N 0102-014-1801 C-25254

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NAVAL ELECTRONIC SYSTEMS COMMAND
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